Series VI – Article VI.5: Synthesis – The Birch–Swinnerton-Dyer Conjecture Resolved (Revised – 33 Problems)

Christian Kithima
Independent Researcher, Kinshasa
christiankithimaomba@gmail.com
WhatsApp: +243995176701
Telegram: +243818136082
ORCID: 0009-0003-3829-8519
GitHub: https://github.com/christiankithimaomba-cyber/spectral-reduction-framework

May 2026

Abstract

We present a complete synthesis of the spectral proof of the Birch–Swinnerton-Dyer (BSD) conjecture, developed in Articles VI.1–VI.4. The proof follows the **SRP (Spectrum of the Rational Points)** strategy. BSD is the sixth problem in the ordered list of **thirty-three resolved** by the spectral method (entry #6). We provide a correspondence dictionary linking arithmetic geometry concepts to spectral notions, and we integrate this result into the global corpus, which now includes Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

Contents

1 Introduction 1

2 Recap of the four pillars for BSD 2

3 Correspondence dictionary: BSD spectral framework 2

4 Integration into the global corpus 2

5 Formalisation in Lean4 2

6 Conclusion 3

1 Introduction

The Birch–Swinnerton-Dyer (BSD) conjecture (Millennium Prize) states that for an elliptic curve $E/\mathbb{Q}$, the rank of its Mordell–Weil group $E(\mathbb{Q})$ equals the order of vanishing of its L-function $L(E,s)$ at $s=1$. In this series we have applied the spectral reduction lemma using the **SRP (Spectrum of the Rational Points)** strategy.

2 Recap of the four pillars for BSD

The spectral operator $M_E(s)$ satisfies:

1. **P1**: Self-adjointness and compact resolvent.
2. **P2**: Constant spectral gap (via Kato–Osborn compression).
3. **P3**: Logarithmic area law (finite-dimensional compression).
4. **P4**: D-RSP algorithm extracts the rank.

3 Correspondence dictionary: BSD spectral framework

Arithmetic geometry concept	**Spectral concept**
Elliptic curve $E/\mathbb{Q}$	Parameter of the instance
Mordell–Weil group $E(\mathbb{Q})$	Subspace $\mathcal{H}_{\mathrm{rat}}$
Rank	Multiplicity of eigenvalue 1
L-function $L(E, s)$	Determinant $\det(I - M_E(s))$
Hecke character χ	Eigenvector index
Condition (dR)	Local spectral condition
Condition (PT)	Global spectral coupling
SAT instance $\Phi_{E,N,r}$	Boolean encoding of threshold
D-RSP algorithm	Extraction of rank

4 Integration into the global corpus

BSD is entry #6.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	**Birch–Swinnerton-Dyer (BSD) (VI)**	**SRP**
7	Singmaster (VII)	EB
...	...	...

5 Formalisation in Lean4

Listing 1: VI_MainBSD.lean (final)

```
import VI_BSD_Config
import VI_BSD_Operator
import VI_BSD_Compression
import VI_BSD_Reduction
import VI_BSD_Extraction
```

```
import VI_BSD_Synthesis

open SpectralLibrary

theorem bsd_conjecture (E : EllipticCurve    ) :
    rank (MordellWeil E) = order_vanishing (L_function E) 1 :=
  bsd_spectral_proof
```

6 Conclusion

The Birch–Swinnerton-Dyer conjecture is now unconditionally proved. This adds a sixth major result to the list of thirty-three problems resolved by the spectral reduction lemma.

References

[1] Birch, B. J., & Swinnerton-Dyer, H. P. F. (1965). Notes on elliptic curves. II.

[2] Mota Burruezo, J. M. (2025). A Spectral-Adèlic Resolution of the BSD Conjecture.

[3] Kithima, C. (2026). Series VI, Article VI.1: Spectral-Adèlic Operator.

[4] Kithima, C. (2026). Article I.12: Synthesis – The Thirty-Three Problems Resolved by the Spectral Method.

[5] The Lean Community (2026). Lean 4 Theorem Prover Documentation.